Filling In Missing Stopping Reasons for Clinical Trials

CS5785 / ORIE5750 / ECE5414 — Midterm Project, Part 1

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Abstract

We predict why a clinical trial stopped early. Each trial can have any of five reasons, and most of the training reasons are hidden. The useful input is a 150-number summary of the stopping-reason text. Eligibility criteria, disease names, drug fingerprints, pathways, and start year did not improve the score, so the submitted model uses the embeddings alone.

The model is a blend of logistic regression, cosine nearest neighbors, and shrinkage linear discriminant analysis. A 30-component Gaussian mixture does not name the reasons. It only helps decide which unlabeled trials are safe to learn from. Those trials are added in two rounds, at a quarter of the weight of a real label. On the 78 validation trials this scores 0.891 sample-wise F1. After a final fit on all 235 real labels, the reported Kaggle score is 0.89636.

Team responsibilities. All three authors worked on Part 1, Part 2, and the report. **Michael Mazzola** led the Part 1 model. **Tarun Mandalapu** built the first Part 2 classifier, which **Anirudha Hadagali** then turned into the final Part 2 ensemble. Tarun Mandalapu wrote the combined project report. This Part 1 write-up follows that report and the later model checks.

1. Methods

1.1. What we are predicting

The data come from ClinicalTrials.gov through the HINT benchmark, with further processing by the course staff [2, 8]. There are 1,561 trials that stopped early. The training file has 1,171 of them. Only 157 have visible reason labels: the earliest 10% by start year, plus 40 extra trials chosen so the rare reasons are not missing. The other 1,014 training trials have every reason marked -100. That value means "unknown," not "this was not a reason." The validation file has 78 fully labeled trials. The test file has 312 later trials, from 2010 to 2016, with no labels.

A trial can stop for more than one reason. The five labels are Safety, Funding, Efficacy, Recruitment, and Other. Among the 235 trials with real labels, Recruitment appears on 64%, Other on 15%, Efficacy on 13%, Safety on 11%, and Funding on 5%. The labeled trials are mostly early, and the test trials are the latest, so these rates will not match the test set. Recruitment is more common early, and Safety is more common later.

The score we care about is sample-wise F1: F1 computed for each trial, then averaged over trials. We also report F1 for each reason and the macro average, which is the plain average of those five F1 scores. One validation trial moves sample-wise F1 by about 1.3 points, so a small gap on 78 trials is weak evidence. Every comparison below was therefore checked a second way, with the same saved grouped cross-validation: eight repeats of five folds on the 157 training labels, and the same plan on all 235 labeled trials. We also scored three time-ordered windows, 98 held-out trials in total. Where a single validation number and the repeated checks disagree, we follow the repeated checks.

1.2. Which inputs we kept

Each trial has a 150-number embedding of its stopping-reason text. The original text is not in the files. The embedding was reduced and randomly rotated, so a single coordinate has no meaning, and the values are small. We standardize them with a scaler fit only on the rows the current split is allowed to learn from. Trial IDs are used to line up rows. They are never used as inputs.

We tried the other fields and then left them out. Start year, eligibility text, disease names, drug pathways, and Morgan fingerprints of the drug structure [3, 4, 6] did not raise the validation score. The reason is simple. The embedding is already a summary of the text the labels were written from. The other fields describe the trial, not why it stopped. A trial can run out of money in any disease. Adding them also means adding many new inputs to a model that starts with only 157 real labels. About 44% of trials have no matched drug, so a fingerprint block is all zeros for those rows. Year is a special trap: it fits the early labeled trials and then has to predict later ones. Section 2 gives the numbers.

1.3. How missing labels are created

The assignment asks for at least one unsupervised method to fill in missing labels. We tried three. k-means and DBSCAN were useful checks. A careful DBSCAN labeling, followed by logistic regression, reached about 0.910 sample-wise F1 on the 78 validation trials in an early comparison. We did not submit it. That number came from a different training setup, and it did not win the later grouped checks against the model below. On its own, clustering mostly separates Recruitment from everything else, because Recruitment dominates the data.

What we kept is a Gaussian mixture [1, 5]. After standardization we keep 30 principal components, about 77% of the variance, and fit 30 diagonal-covariance components. The components are started from k-means++ assignments. We run three restarts from `random_state` $= 0$ and keep the restart with the best log-likelihood. Expectation–maximization stops when the average log-likelihood improves by less than 10^{-3}, or after 100 iterations. The kept run converged in 29 iterations. The mixture is fit only on rows the current split may see. Test rows are not included.

Each component is turned into a reason profile by averaging the real labels of the trials assigned to it, weighted by how strongly the mixture assigns each trial. A trial with two reasons contributes half a count to each, so every trial has the same total weight. We add one pseudo-count of the overall reason rates. That stops a component with few labeled trials from collapsing to a single reason. The mixture's guess for a new trial is the average of these profiles, weighted by the trial's component memberships.

We do not trust that guess as a label. In the teaching step it has only a 5% vote. The classifier blend below has the other 95%.

1.4. The classifier

The submitted classifier is one logistic regression over the five reasons ($C = 0.1$, at most 3,000 iterations, tolerance 10^{-4}), not five separate classifiers. A trial with two reasons is entered once for each reason, and one unit of weight is split evenly between those copies. The class weights are computed from the real labels only, then frozen. Rare reasons are upweighted so Recruitment does not drown them out. A label the model made up later cannot change those weights. A made-up row also counts as only a quarter of a real row.

We average this logistic model with two others, using fixed weights. Cosine nearest neighbors get 30%. Shrinkage linear discriminant analysis gets 20%. Logistic regression gets the remaining 50%. The neighbor model looks at the 10 closest trials and weights them by the square of the cosine similarity, so a close match counts for much more than a distant one. A neighbor that has two reasons votes for both. The discriminant analysis uses automatic shrinkage and equal class priors, so it is not pulled toward Recruitment just because that reason is common. All three models see the full 150 standardized embeddings. Only the mixture uses the 30 principal components.

This blend is the later version of our first strong model, with three changes that the grouped checks supported. We froze the class weights on the real labels, we let neighbors vote for every reason they carry, and we cut the mixture's teaching vote from 20% to 5%.

1.5. Teaching the model, then deciding

We add unlabeled trials in two rounds [7, 9]. Each round scores every still-unlabeled trial with the 95/5 blend above. If its highest reason scores at least 0.8, we treat the decision rule below as that trial's label, add the row at quarter weight, and refit. Each round starts again from the real labels plus the rows accepted in that round. It does not keep stacking old guesses. In the final fit, on all 235 real labels, the first round accepts 685 unlabeled trials and the second accepts 757. Their total weight is about 103 and then 120, against 235 for the real labels, so the guesses cannot outvote the data.

The course also asks for a reason label on every originally masked training row. After the final fit we predict all 1,014 of those rows. That file is not the same thing as the 757 rows used during training. The 157 original training labels are copied through unchanged.

Every trial in this corpus stopped for a recorded reason, so a prediction of "no reason" is always wrong. We always keep the highest-scoring reason, and we add any other reason scoring at least 0.3. These scores are for making that choice. They are not calibrated probabilities.

1.6. What the final fit is allowed to see

Model choice used the 78 validation trials, with the model trained on the 157 labeled training trials. The submitted fit is different. It retrains the same fixed recipe on all 235 real labels, and it may use the 1,014 masked training rows for the mixture and for generated labels. The 312 test rows are used only to make the final predictions. Because the validation labels are inside that final fit, the 0.891 validation score is not an independent score of the submitted file.

Our first submitted model, at a reported Kaggle score of 0.89423, had built its scaler and clusters with the test rows included. We stopped doing that. The later models, including the one below, never see a test row until prediction.

2. Results

Table 1 is the first model family. The mixture by itself is the weakest result. One logistic model over all five reasons beats a separate classifier for each reason. Letting the model teach itself is the largest single gain. Giving the mixture a small role in that step gives the best cross-validated score in

Validation		Repeated CV	
samp. F1	macro F1	samp. F1	macro F1
One-vs-rest logistic regression			
0.868	0.706	0.898	0.797
Cosine nearest neighbors (k=10)			
0.872	0.607	0.881	0.734
Mixture used as the labeler			
0.780	0.570	0.819	0.640
Logistic regression, no class weights			
0.850	0.677	0.894	0.765
Balanced logistic regression + neighbors			
0.885	0.724	0.898	0.800
Self-training, mixture left out			
0.889	0.723	0.913	0.805
Self-training with the mixture			
0.893	0.714	0.914	0.814

Table 1. First model family. Validation is the 78 labeled trials. Repeated CV is eight repeats of five folds over all 235 labeled trials. The last row is the model behind the reported 0.89423 submission.

Validation		Repeated CV	
samp. F1	macro F1	samp. F1	macro F1
Embeddings only			
0.893	**0.714**	0.914	0.814
+ *start year*			
0.891	0.711	0.912	0.807
+ *eligibility text*			
0.876	0.709	0.913	0.801
+ *disease names*			
0.876	0.696	0.914	0.806
+ *drug fingerprints*			
0.889	0.719	0.918	0.815

Table 2. Feature check on the first self-training model. No added block improves validation sample-wise F1.

this family. That model, refit on all 235 real labels, is the reported 0.89423 Kaggle submission. As noted above, its preprocessing saw the test rows, so we do not treat it as the same experiment as the later models.

Table 2 adds one extra input block at a time to that same model. None of them raises validation sample-wise F1. Morgan fingerprints move the cross-validated score by 0.004, which is inside the noise between repeats. Later checks that added diseases, pathways, or eligibility text to an embeddings-only logistic model also failed to help, and the pathway block lowered the score. We kept the embeddings alone.

Model	Train	Pooled	Val.	Time	Kaggle
Final blend	**.9321**	.9214	.891	**.884**	**.89636**
LDA ensemble	.9318	**.9216**	**.906**	.878	.89401
Earlier LDA blend	.9281	.9154	.872	.874	—
First submission	.9196	.9098	.891	.861	.89423

Table 3. Later models on the same grouped checks. Train is cross-validation on 157 labels. Pooled is cross-validation on 235 labels. Val. is the original 78 validation trials. Time pools the three chronological windows. Kaggle scores are the reported public scores.

Table 3 compares the later models on one shared set of checks. The final model is the one described in Section 1. On cross-validation it is tied with the linear-discriminant ensemble: 0.9321 against 0.9318 on the 157 training labels, and 0.9214 against 0.9216 on all 235. Its 78-trial validation score is lower, 0.891 against 0.906. Its time-ordered score is higher, 0.884 against 0.878. The reported Kaggle score is 0.89636, above both 0.89423 and the ensemble's 0.89401. We submitted it because that was the best reported test score, and the repeated checks did not show a real loss.

On the 78 validation trials, the final blend scores 0.941 on Safety (9 trials), 0.667 on Funding (4), 0.759 on Efficacy (14), 0.990 on Recruitment (50), and 0.471 on Other (10). The sample-wise average is 0.891 and the macro average is 0.765. The two common reasons are handled well. Funding and Other are not, and each rests on too few trials to trust a small difference. The grouped cross-validation macro F1 is 0.883 on the 157 training labels and 0.850 on all 235.

3. Discussion

The mixture is a poor labeler and a useful tie-breaker. Used on its own, the mixture is the worst row in Table 1. Its components mostly separate Recruitment from the other reasons. Adding more components splits Recruitment into smaller pieces. It does not isolate Safety or Funding. The embeddings do contain those reasons: a supervised model reaches 0.941 F1 on Safety from the same inputs. The problem is the clustering. It finds the largest pattern in the data, and that pattern is Recruitment.

The mixture still earns its place once it is demoted. In the first family, using it as a 20% voice in self-training raised cross-validated macro F1 from 0.805 to 0.814. In the submitted model we cut that voice to 5%. Its job is to steady which unlabeled trials get promoted, not to name their reasons. The quarter weight on a generated row does the same kind of work. In the final fit the 757 accepted rows carry about half the weight of the 235 real labels, so a confident guess can help and cannot take over.

Why adding every feature makes the model worse. This was the clearest negative result. The embeddings are a compressed copy of the stopping-reason text, which is the text the labels came from. They already hold most of what the labels have to say. Criteria, diseases, and drugs describe what the trial was, not why it ended. Adding one of those blocks means 20 to 2,000 new inputs on 157 real labels. The noise shows up immediately. Eligibility text drops validation sample-wise F1 from 0.893 to 0.876 in Table 2. Fingerprints barely move the cross-validated score, and 44% of trials have no matched drug. Start year looks predictive only because the labeled trials were chosen to be early. The test trials are the late ones, so a year pattern learned here points the wrong way. Leaving a feature out was a result, not a shortcut.

A higher validation score was not enough. The linear-discriminant ensemble beats the submitted blend by 0.015 on the 78 validation trials and is tied with it under cross-validation. We did not follow the validation number. Four Funding trials and ten Other trials can move the macro average by several points, which is why the validation and cross-validated columns in Table 1 already disagree about which early model is best. The time-ordered windows, which look more like the later test years, favor the submitted blend. So does the reported Kaggle score. The validation score also belongs to a fit that has not yet seen those 78 labels. The submitted file is retrained with them included.

What we would not do again. We would not fit the scaler or the mixture on the test rows. That choice is part of the 0.89423 submission, and the later inductive fit scored higher. We would not let generated labels rewrite the class weights, because the model then starts believing its own class balance. We would not add year, text, or drug features in the hope that more inputs are safer. The gain that held up was smaller than that search: freeze the weights, keep both of a neighbor's reasons, and let the mixture advise instead of decide.

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